Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 233 (2024) iii-viii

Procedia Computer Science

www.elsevier.com/locate/procedia

Table of Contents

Secure Goods Storage and Anti-Theft Approach using Ethereum Blockchain
Likhitha Amasala, Mahesh Ponnuru, and Sridevipomalar P 1

An Enhanced Dynamic Multi-Path Routing Algorithm for Software Defined Networks
Venkata Sai R., Prabu U., Satwik Ch., Sanjaya Kumar Panda, and Geetha V. 12

Integrated Leaf Disease Recognition Across Diverse Crops through Transfer Learning
Utkarsh Yashwant Tambe, Dr. A. Shanthini, and Pao-Ann Hsiung 22

A Pre-Trained Model for Aspect-based Sentiment Analysis Task: using Online Social Networking
Amit Chauhan, Aman Sharma, and Rajni Mohana 35

Insightful Clinical Assistance for Anemia Prediction with Data Analysis and Explainable AI
E Kasthuri, S. Subbulakshmi, and Rajasree Sreedharan 45

Solar Irradiance Forecasting using Improved Sample Convolution and Interactive learning
Ansil Subair, and Gopakumar G 56

Revolutionizing Missing Data Handling with RFKFCM: Random Forest-based Kernelized Fuzzy C-Means
Jyoti, Jaspreeti Singh, and Anjana Gosain 66

An Effective Deployment of Controllers in Software-Defined Networks
Sk Yasmine, U Prabu, Y Sri Deepak Phaneendra, and V Geetha. 77

A Novel Architecture for Community Detection Between Large Social Media Creators
Dhruvjyoti Swain, S Eesha, Gokul D. Raj, and T. Anjali 87

Enhancing Veracity: Empirical Evaluation of Fake News Detection Techniques
KS Sreekar Datta, G Narasimha Naidu, S Abhishek, and Anjali T 97

IndEmoVis: A Multimodal Repository for In-Depth Emotion Analysis in Conversational Contexts
Ruhina Karani, and Dr. Sharmishta Desai 108

Privacy in Smart Homes with Remote User Authenticated Key Establishment Protocol
Neha Sharma, and Pankaj Dhiman. 119

Prediction of Consultation Wait Time in Outpatient Clinic: An Approach using Neural Network with Optimized Feature Selection
Jeffin Joseph, S. Senith, A. Alfred Kirubaraj, and S.R. Jino Ramson 129

A Multithreaded-BFS Algorithm for Optimizing the Graph Computation in Multicore Processing System
Kapil Juneja, and Dhiraj Khurana 138

Analysis of Block Chain Applications in Asset Backed Securitization System
Ashish Kumar, Rohit Kumar Sachan, Divya Singh, and Rachna Jain. 147

Machine Learning Approach for Predicting the Net Asset Value (NAV) of Mutual Funds based on Portfolio Holdings
Rohit Kumar Sachan, Shabanam Kumari, Vipul Khandelwal, and Tarun Kumar 154

Attention-Based AdaptSepCX Network for Effective Student Action Recognition in Online Learning
Arnab Dey, Anubhav Anand, Subhajit Samanta, Bijay Kumar Sah, and Samit Biswas 164

DDoS Attack Detection Model using Machine Learning Algorithm in Next Generation Firewall
Manthan Patel, P.P. Amritha, Vinay B Sudheer, and M. Sethumadhavan 175

Echoes of Truth: Unraveling Homophily in Attributed Networks for Rumor Detection
S.V. Rithish, C.R. Prabu, M.B. Anuush, and L.R. Deepthi 184

Immutable COVID-19 Vaccination Certificate using Blockchain
Abdul Muqsit Haji Jafari, Ravi Kumar Patchmuthu, and Sharul Tazrajiman Haji Tajuddin. 194

Enhancement of Multiclass Dermoscopic Image Classification with a Fusion Model and Fine-Tuned Deep Neural Networks
Sonam Khattar, and Dr. Rohit Bajaj. 204

Optimizing Speaker Identification through SincsquareNet and SincNet Fusion with Attention Mechanism
Banala Saritha, Rabul Hussain Laskar, Madhuchhanda Choudhury, and Anish Monsley K. 215

Segmentation and Classification of Diabetic Retinopathy using Ensemble Deep Neural Network
Anju Mishra, Mrinal Pandey, and Laxman Singh 226

Analysis of Lightweight Cryptographic Algorithms for IoT Gateways
Ch. Jnana Ramakrishna, D. Bharath Kalyan Reddy, B.K Priya, P.P Amritha, and K.V Lakshmy 235

Renewable Energy Trading Platform using Blockchain
Aditya Shinde, Vijay Mane, Nahush Ambulgekar, and Zayd Ansari. 243

Artificial Intelligence and Deep Learning-based Model for Indoor Environment: with Virtual Reference Tag Allocation
N.memathavan@gmail.com Mathavan, and Dr. S. Sivaranjani. 254

Silent Expressions Unveiled: Deep Learning for British and American Sign Language Detection
Sai Niketh Koyineni, Gurram Kumar Sai, Kalwa Anvesh, and T Anjali. 269

Medical ChatBot Assistance for Primary Clinical Guidance using Machine Learning Techniques
Sreedhar Kumar S, Syed Thouheed Ahmed, Afifa Salsabil Fathima, Nishabai M, and Sophia S. 279

Diagnosing Diabetes using Machine Learning-based Predictive Models
D Kaviyaadharshani, M Nivedhidha, R Jeyarohini, J Lece Elizabeth Rani, M P Ramkumar, and G S R Emil Selvan 288

The Analysis of Neural Network Models to Distinguish AI generated faces from Real faces
Joshita Malla, Harshini Vemuri, SreeDivya Nagalli, S Abhishek, and T Anjali 295

Predicting Hepatocellular Carcinoma Graft Survival Rate in Post Liver Transplantation Using DeepHit
Devi Rajeev, S Dr. Remya, Dr. Anand Nayyar, and Dr. Krishnanunni Nair 307

Deep Learning based Pulmonary Embolism Detection using Convolutional Feature Maps of CT Pulmonary Angiography Images
Sanjana Sukumar, A. Harish, A. Shahina, Sanjana B, and A. Nayeemulla Khan 317

Enhancing Village Ranking: Leveraging Cluster Analysis and Machine Learning
Moturi Karthikeya, S Madhan, Akhbar Sha, Megha R, Dhanush Krishna R, and G. Gopakumar 327

Data-Driven Clustering and Insights for Rural Development in India
Akhbar Sha, S Madhan, Moturi Karthikeya, Megha R, Dhruvjyoti Swain, and G. Gopakumar. 336

A Novel Predictive Analysis and Classification of Land Subsidence Vulnerability Mapping based on GIS using Hybrid Optimized Machine Learning Techniques and Computer Vision
Senthil G. A, R. Prabha, Arun M, A. Sahaya Anselin Nisha, and S. Prabu 343

Sign Language Recognition by using Spatio-Temporal Features
Renjith S, Manazhy Rashmi, and Sumi Suresh M S 353

Convolutional Neural Network-Enhanced Video Artistry: Leveraging VGG16 for Dynamic Style Transfer
Tharun S Kumar, Nadeem N, Siddharth Rajesh Menon, and T. Anjali 363

Biotic Stress Management in Soil-Less Agriculture Systems: A Deep Learning Approach for Identification of Leaf Miner Pest Infestation
A. Subeesh, and Naveen Chauhan 371

From Posts to Personas: Classifying Users in the Social Media Landscape
Lekshmi S Nair, and Jo Cheriyan. 381

Phonogram-based Automatic Typo Correction in Malayalam Social Media Comments
D Jyothi Ratnam, A.N Karthika, K Praveena, Rajesh Taniya, S Thara, and Nedungadi Prema 391

Hybrid Actor-Action Relation Extraction: A Machine Learning Approach
Reshma P. Nair, and M.G. Thushara 401

CNN Fusion: A Promising Technique for Ophthalmic Disorder Diagnosis
Ankur Biswas, and Rita Banik 411

Detection and Classification of Respiratory Syndromes in Original and modified DCGAN Augmented Neonatal Infrared Datasets
S Sarath, and Jyothisha J Nair 422

ShardedScale: Empowering Blockchain Transaction Scalability with Scalable Block Consensus
Praveen M Dhulavvagol, S G Totad, Atrey Mahadev Anagal, Swaroop Anegundi, Praveen Devadkar, and Vinayak Sudhakar Kone. 432

A Deep Learning-based Transfer Learning Approach Fine-Tuned for Detecting Diabetic Retinopathy
Devendra Singh, and Dinesh C. Dobhal 444

Rumor Source Detection in Subgraphs: An ML Approach
L.R. Deepthi, and S Nair Lekshmi. 454

A Deep Learning-based Smart System for Ornaments and Devices Detection during Check-In of Entrance Exams
P S Krishnapriya, Gundam Nikitha, Kalla Vishwas Uday Kiran, Kuruba Aparna, and Singamsetty Chandu Priya 464

Automatic Detection of Fake News Using Gated Recurrent Unit Deep Model
Hanane ELFAIK, and El Habib NFAOUI 474

EDTBERT: Event Detection and Tracking in Twitter using Graph Clustering and Pre-trained Language Model
Abhaya Kumar Pradhan, Hrushikesha Mohanty, and Rajendra Prasad Lal. 481

CNN-LSTM Hybrid Model for Enhanced Malware Analysis and Detection
Gautam Karat, Jinesh M. Kannimoola, Namrata Nair, Anu Vazhayil, Sujadevi V G, and Prabaharan Poornachandran. 492

A Comparative Study Of Super-Resolution Interpolation Techniques: Insights For Selecting The Most Appropriate Method
Maganti Jahnavi, D. Rajeswara Rao, and Amballa Sujatha. 504

Unveiling Underwater Structures: MobileNet vs. EfficientNet in Sonar Image Detection
P A Arjun, S Suryanarayan, R S Viswamanav, S Abhishek, and T Anjali 518

Bidirectional Gated Recurrent Unit with Glove Embedding and Attention Mechanism for Movie Review Classification
M Greeshma, and Philomina Simon 528

Precaution Chatbot for Epilepsy Patients using Natural Language Processing and Deep Learning Sequential Model
E Kasthuri, S. Subbulakshmi, and Rajasree Sreedharan 537

Enhancing Cross-Linguistic Image Caption Generation with Indian Multilingual Voice Interfaces using Deep Learning Techniques
Vijay A Sangolgi, Mithun B Patil, Shubham S Vidap, Satyam S Doijode, Swayam Y Mulmane, and Aditya S Vadaje 547

Implementation of Efficient Machine Learning Techniques for Prediction of Cardiac Disease using SMOTE
Narayanan, Jayashree 558

Checking Counterfeit Critiques on Commodities using Ensemble Classifiers Enhancing Information Credibility
Ram Chatterjee, Mrinal Pandey, Hardeo Kumar Thakur, and Anand Gupta 570

A Novel Approach for Bluetooth Mesh Network Detection using Machine Learning and Correlation Analysis
Ravi Kiran Varma Penmatsa, Krishna Kanth Varma Penmatsa, Leela Sai Pavan Kethinedi, Dheeraj Sakkuri, and Rama Krishna Sai Satwik Ramineedi 580

Student's Interest and Opinion Towards Online Education
Hemanth Sravan Kumar Veguru, J Naren, and Yasasree Singam 590

Bird Sound Identification System using Deep Learning
Indumathi C P, Diviyalakshmi K R, and Mahalakshmi R 597

Optimizing Task Scheduling in Cloud Computing: An Enhanced Shortest Job First Algorithm
Yellamma Pachipala, Kavya Sri Sureddy, A.B.S. Sriya Kaitepalli, Nagalakshmi Pagadala, Sai Satwik Nalabothu, and Mihir Iniganti 604

Investigation on Preserving Privacy of Electronic Medical Record using Split Learning
Kiruthika M, Abhishek Kumar, Lalitha Krishnasamy, and Velliangiri Sarveshwaran. 614

Flask-based ASR for Automated Disorder Speech Recognition
Devalla Bhaskar Ganesh, Yellama Pachipala, Syed Sania Rizvi, Teena Chowdary Manne, Himavanth Swamy Atchi, and V V R Maheswara Rao 623

Design of a Multiple Phased Convolution Model for Pre-emption of Breast Cancer via Extensive Learning Operations (MPC2EL)
Pravin Malve, and Dr. Vijay Gulhane 638

Animal Identity Recognition using Object Detection Techniques
Mohd Zeeshan Ansari, Faiyaz Ahmad, Erfan Naseri Taheri, Ravi Gowri Jaswanth Reddy, and Faria Mabood 651

An Empirical Investigation of Docker Sockets for Privilege Escalation and Defensive Strategies
jyashree Ra, Senthilkumar Mathi, Saravanan G, and Sakthivel M 660

Comparative Analysis of Various Feature Extraction Methods on IoT 2023
M Vergin Raja Sarobin, J Ranjith, D Ashwath, K Vinithi, Smiti, and V Khushi 670

Transformer Based Sentiment Analysis on Code Mixed Data
Koyyalagunta Krishna Sampath, and M. Supriya 682

Data Biasing Removal with Blockchain and Crowd Annotation
Avijit Bose, Dr. Pradyut Sarkar, and Dr. Premananda Jana 692

Improvement of Brain Tumor Categorization using Deep Learning: A Comprehensive Investigation and Comparative Analysis
T. Lakshmi Prasanthi, and N. Neelima 703

Prioritization of Risks in Agile Software Projects Through an Analytic Hierarchy Process Approach
Ayesha Ziana M, and Charles J 713

A Comprehensive Performance Analysis of a Hyperledger Fabric-powered Blockchain Network for Cross-Border Fund Transfers
Pradnya B. Patil, and M. Sangeetha 723

Real-Time Speaker Identification and Subtitle Overlay with Multithreaded Audio Video Processing
Sahith Madamanchi, Gona Kushal, Srikesh Ravikumar, Puli Dhanvin, Remya M S, and Prema Nedungadi 733

Automated Classification and Size Estimation of Fetal Ventriculomegaly from MRI Images: A Comparative Study of Deep Learning Segmentation Approaches
K Gopikrishna, N R Niranjan, Siddharth Maurya, V G Uday Krishnan, and Simi Surendran 743

Performance Analysis of Stocks using Deep Learning Models
Danish Raza Rizvi, and Mohd Khalid 753

Fake News Detection using Python and Machine Learning
Jumana Jouhar, Anju Pratap, Neharin Tijo, and Meenakshi Mony 763

Towards Sustainable Farming: Leveraging AIoT for Precision Water Management and Crop Yield Optimization
Shashank Karn, Radhika Kotecha, and Ritesh Kumar Pandey 772

Study on Limited Data Problem - Semantic Segmentation of Retinal Fundus Images
Uday A Nuli, Shrinivas D Desai, and Gururaj N Bhadrid 782

Traffic Management through Cutting-Edge Vehicle Detection, Recognition, and Tracking Innovations
S Kanagamalliga, P Kovalan, K Kiran, and S Rajalingam 793

Variational Auto encoders for Improved Breast Cancer Classification
V Sreelekshmi, and J Nair Jyothisha 801

BiT5: A Bidirectional NLP Approach for Advanced Vulnerability Detection in Codebase
Prabith GS, Rohit Narayanan M, Arya A, Aneesh Nadh R, and Binu PK 812

Feedback-Regulated Information Fusion Approach for Optimizing Multiple Spread Control in complex Network
Jo Cheriyan, and Jyothisha J Nair 822

Automated Bone Fracture Detection in X-ray Imaging to Improve Orthopaedic Diagnostics in Healthcare
Soumi GhoshJahiruddin Mallick, Monalisa Chakraborty, and Subir Gupta 832

BEE Friendly Flora: Comparative Analysis of Plant Preference among Native and Non-Native Bee Species using EDA and Machine Learning Model
Praveen M. Dhulavvagol, S Pritam, A Vijayalaxmi, S Raghav, and B Sneha 841

Parallel VLSI Architectures for High-Throughput Image Processing in Medical Imaging
P Vinayagam, R Yuvaraaj, S Sathiyanandham, and S Kanagamalliga. 851

Enabling Attribute-based Access Control for OpenStack Cloud Resources through Smart Contracts
Ganesh Dalabanjan, and Narayan D G 861

Enhancing Drug Recommendations: A Modified LSTM Approach in Intelligent Deep Learning Systems
S Siji Rani, P Shilpa, and Aswin G Menon 872

Deep Learning-based Cavity Detection in Diverse Intraoral Images: A Web-based Tool for Accessible Dental Care
Siji Rani S, Srija Garine, Papolu Hema Janardhana, Lakkireddy Lakshmi Prabhanjan Reddy, Penubothu Jagadeesh Venkata Kumar, and Chapa Gagan Dwaz. 882

Temporal Fusion of Time-Distributed VGG-16 and LSTM for Precise Action Recognition in Video Sequences
K A Athira, and J Divya Udayan 892

Evaluating the Effect of Preprocessing Tools for Marathi Text Retrieval
Harshali B. Patil, and Ajay S. Patil 902

Machine Learning as a Service Cloud Selection: An MCDM Approach for Optimal Decision Making
Seema Gupta Bhol, Satarupa Mohanty, and Prasant Kumar Pattnaik 909

WebGL vs. WebGPU: A Performance Analysis for Web 3.0
Satyadhyan Chickerur, Sankalp Balannavar, Pranali Hongekar, Aditi Prerna, and Soumya Jituri 919

Design and Implementation of Artificial Neural Network Classifiers based on Hypertuning Parameters for Breast Cancer Diagnosis
Neetu Gupta, Hemant Kumar Gupta, Rahul Srivastava, Chhavi Saxena, Surjeet 929

Traffic Sign Classification using Deep Learning Comparative Study
ASSEMLALI Hamza, and SAEL Nawal. 939

YOLOv5x-based Brain Tumor Detection for Healthcare Applications
Manoj Kumar, Urmila Pilania, Stuti Thakur, and Tanisha Bhayana 950

Towards Detection of Network Anomalies using Machine Learning Algorithms on the NSL-KDD Benchmark Datasets
Amol D. Vibhute, Chandrashekhar H. Patil, Arjun V. Mane, and Karbhari V. Kale 960

An Ensemble Classification Model for Phishing Mail Detection
Anirudh S, P Radha Nishant, Sanjay Baitha, and K Dinesh Kumar 970

Satellite Image Classification using CNN with Particle Swarm Optimization Classifier
Vidhya S, Balaji M, and Kamaraj V. 979

An Innovative Novel Transformer Model and Datasets for Safeguarding Religious Sensitivities in Online Social Platforms
Mohammad Shariful Islam, Mohammad Abu Tareq Rony, Mejbah Ahammad, Shah Md Nazmul Alam, and Md Saifur Rahman 988

Identification of Kidney Disorders in Decentralized Healthcare Systems through Federated Transfer Learning
Varun Vekaria, Raj Gandhi, Bhargavee Chavarkar, Hetvi Shah, Chetashri Bhadane, and Poonam Chaudhari . . . 998

LAKBAYAN: A Travel-Social Itinerary Application with Derived Enhanced Simulated Annealing
Julia Coleene B. Panggoy, Nexxus Jude V. Sealana, Rhodessa J. Cascaro, Patrick D. Cerna, and Patricia Anne Marie D. Go 1011

Explainable AI in Deep Learning Based Classification of Fetal Ultrasound Image Planes
Akshay Harikumar, Simi Surendran, and S Gargi 1023

Spectral Segmentation Augmented with Normalized Cuts for Detection of Early Blight Disease in Potato
Deepkiran, Mrinal Pandey, and Laxman Singh 1034